University of North Florida
Calculus-Based Physics II CCEC PHY 2049 (CRN:84245)
Fall 2026

Course Syllabus

This document is the syllabus for Calculus Physics 1I (PHY 2049), it is the controlling document for the class therefore you are responsible for knowing all the information presented in this document and doing what is required of you for this class. If you have any questions about what is contained in this syllabus, please contact the instructor so that they can answer, explain or clarify any misunderstandings. Below you will find most of the information you need to be successful in this course.

Class Information

Instructor: Mr. Jim Montgomery **Office #:** 2102 (2nd floor Building #50)

e-mail: j.montgomery.138251@unf.edu

Classroom: Building 50 room 1102 **Class Time**: Monday Wednesday and Friday from 8:00 to 8:50 am

- Note: This class meets at 8:00 am Monday, Wednesday and Friday, if you are not willing to make the commitment to attend this class time please reconsider taking this class section!

Office Hours in person or by Zoom Appointment:

Tuesday: 1:00 p.m. to 3:00 p.m.
Wednesday: 10:00 a.m. to 12:00 p.m.
Thursday: 10:00 a.m. to 3:00 p.m.
e-mail for an appointment or a different day and time.

Required Technology: It is mandatory for each student to have access to a laptop, tablet, or computer. Students must be able to view materials on Canvas such as lecture notes (pdf files), quizzes, and videos. You will also need some method of scanning multi-page documents for uploading to Canvas as one PDF file. Only having a cell phone for this class will not work.

Required Calculator: Any Scientific calculator (Standard or graphing calculators are acceptable).

Textbook: Lecture notes posted in the Canvas Modules are the class textbook, there is an optional free online textbook for reference **"University Physics" OpenStax**, This book is for reference only and is available in web view and PDF for **free**. You can also purchase on iBooks or get a print version, if you prefer from seller OpenStax on Amazon.com.

Free University Physics Volume 2 Book for Download - OpenStax
Free University Physics Volume 3 Book for Download - OpenStax

Overview: This introductory course in physics will investigate the world around us, the students will use an understanding of the Laws of Nature and the use Critical thinking skills to describe this world. In this class the students will acquire knowledge of current scientific understanding of the history and nature of the universe.

Topics include Electric Charge, Electric and Magnetic forces and Fields, Electric circuits, EM induction, EM waves, and Optics.

General Education: This course touches on several areas of importance for general education by: Demonstration a general knowledge of current scientific understanding of the history and nature of the universe, demonstration a general knowledge of the methods and traditions of analysis in natural sciences, use systematic processes, including the collection and analysis of evidence to form and support conclusions, demonstrating proficiency in solving problems using mathematical concepts and quantitative reasoning, demonstrating a general knowledge of the nature, origins, and contributions of major civilizations.

Learning Objectives – after the successful completion of this class you should be able to:

CO 1: Recognize fundamental terms, concepts, principles, and processes in Physics, including how physics relates to other scientific disciplines and the natural world around us.

CO 2: Apply the appropriate mathematical and computational models and methods to solve problems and evaluate the results.

CO 3: Represent and manipulate vectors to solve physics problems using algebra and calculus.

CO 4: Demonstrate a knowledge of "fields" in physics and understand the source of electric and magnetic fields.

CO 5: Apply Coulomb's Law to calculate electric forces and fields for point charges and continuous charge distributions.

CO 6: Use Gauss's Law to determine electric fields in systems with spherical, cylindrical, and planar symmetry.

CO 7: Relate electric potential to the electric field and compute potential energy in electrostatic systems.

CO 8: Analyze Resistor, capacitors and inductors in series and parallel, and calculate energy stored in electric fields.

CO 9: Solve DC circuit problems using Ohm's Law and Kirchhoff's Rules.

CO 10: Calculate magnetic fields using the Biot–Savart Law and Ampère's Law for current-carrying conductors.

CO 11: Determine magnetic forces on moving charges and current-carrying wires.

CO 12: Apply Faraday's Law and Lenz's Law to analyze electromagnetic induction and their applications.

CO 13: Analyze AC circuits involving resistors, capacitors, and inductors using phasor techniques and impedance.

CO 14: Describe reflection and refraction of light both in words and mathematically, explain the phenomenon associated with each.

CO 15: Use the lens and mirror equations along with ray tracing to determine image location, magnification, and type of image.

CO 16: Explain wave optics phenomena, including interference, diffraction, and polarization.

Class Structure

Class structure: This class is designed to be in-person lecture style class with three weekly class meetings on Monday, Wednesday and Friday. This is a "Flipped" class therefore there will be lecture notes to read, quizzes and videos to watch before coming to class.

- Lecture videos and class lecture notes along with a short reading quiz will be posted on Canvas, in the Modules section and should be viewed and read before coming to class.
- There will be a lecture, and "In-class" activities that students will work on in small groups. It will be helpful to bring to class: pencil, paper and calculator for these activities.
- Computer simulations assigned as Interactive lessons will be used in this class to develop a deeper understanding of the Physics we are covering and to see it in a "real world" setting.
- There will be take home worksheets for further practice that will be uploaded to Canvas for grading.

- **Exams will be administered in-class on Campus, there will be NO online exams!**

What is a flipped Class? The student starts by learning the basic course content at home by watching videos and completing activities online, then when they come to class, they get to practice using what they have learned. The class time is dedicated to teaching concepts and problems that are more difficult for the student, with time to answer questions, in-class activities and do demonstrations that enhance the learning. The instructor will be acting as your guide through Physics; the instructor will be there to help you master the concepts and clear up misconceptions.

- It is Important with a Flipped Class to **keep up**, read the lecture notes, watch the videos and complete the activities before coming to class, the instructor will assume you have a basic knowledge of the material and will NOT cover what has been covered in the activities!
- You should set aside a minimum of **3 hours per week to just prepare for class**, this means watching the Lecture videos, reading the lecture notes, and reviewing lecture notes/problems and **6 more hours studying the material**.

Class sessions: Attendance to lectures will be taken and is 5 % or your overall grade. A great deal of the learning in this course occurs during lectures through examples, demonstrations and hearing concepts explained. If you must miss a lecture, it is your responsibility to make sure you find out the information that you missed. Please remember to turn cell phones and other electronics off during class unless they are being used for the class.

The classes will generally be lectures and discussions on the topics listed in the attached schedule. Problems that illustrate the subject being discussed will be solved during the lectures. You should study the assigned text material before the period in which it will be discussed so that you will be prepared to ask questions during class about those concepts you do not understand. Questions and discussion are encouraged during class but may have to be limited due to time constraints, if you have questions that are not answered in class please get help after class from the instructor, lab instructor, or tutor (see Help section).

Every effort will be made to cover all topics outlined on the syllabus during the class, but time may not allow this to be accomplished. **Even though the material may not be covered during the lecture you are still responsible for knowing the content of the material outlined in this syllabus!**

Things to help you in this class

Help with Canvas: Please use this link to self-enroll in ***SOLO: Student Online Learning Orientation course***
This course will help you identify strategies to be successful in the online environment and in using Canvas. If you have never self-enrolled in a Canvas course, detailed instructions are available here. You only have to complete SOLO once, so if you completed SOLO previously for another course, you don't need to repeat it for this course.

Class-time Notes: The instructor will be using "OneNote" for Windows with a tablet to write all notes during class time. These notes can be accessed through OneNote; an e-mail will be sent at the start of the semester with a link to the class notes and posted on Canvas. Review the class notes after class and study how problems were solved.

Lecture notes: Lecture notes will be provided on Canvas for each chapter covered, they can be printed, used for notes while watching the videos and used for review. **The lecture notes have the material and course content you will be responsible for in order to master the learning objects in this course.** The lecture notes are the principal material for this class and determines what you need to know for the assessments!

Practice Problems: Practice problems will be posted for each chapter that is covered, the practice problems will have sample problems and concept questions very similar to what will be on the exam. It is **STRONLY RECOMMENDED** that you work through the problems, starting with an area where you feel a need to re-enforce and concept questions. The practice problems are a great resource for your study group to work through.

Problem Solving Videos: The instructor will post videos of worked out solutions for a sample of the more challenging problems on Canvas to demonstrate the proper methods of problem solving. The propose of the Practice Problem videos are to model the proper method of problem solving and to demonstrate the conceptual and mathematical methods that are being used. To acquire the most benefit from this activity the student should attempt to solve the problems first then review the video for help and to ensure the proper methods of problem solving.

Class Activities: A worksheet with problems will be solved by the whole class and these problems will be worked out in small groups. The Class Activities will be posted on Canvas on the day of the class along with solutions posted after class.

Why attend class and participate in the class activities?

1) The activities help you understand concepts, work problems with help from your neighbors and the instructor, you are not alone in the process, and you can get feedback quickly.
2) You learn what you don't know and can work more efficiently at learning about all the topics covered.
3) You get to apply what you know and struggle with what you don't without hurting your grade.
4) You can ask and get answers to the silly or basic questions that will stump you at home or during the exam.
5) You learn the material with help of others and by helping others, form connections with fellow classmates and find members for study groups outside of class.
6) It is good experience for taking the exams, you get to attempt problems "cold" and learn how to solve them.

Help: Help is available from your instructor during office hours, or you can e-mail and set up a time for a meeting. There are free tutors in the Student Academic Success Services (SASS) and Society of Physics Students. I will also make available solutions to worksheets, class activities and practice problems on Canvas.

Tutoring at UNF: Student Academic Success Services (SASS) is offering free tutoring for this course throughout the semester. Check out their live tutoring schedule on the UNF website: UNF: Live Schedule for Tutoring. Tutoring is free and you can attend as many sessions as you need to get assistance or simply go and practice your skills.

Zero online solutions Policy: This class has a zero browsing for solutions policy which includes websites such as any AI sites, Chegg or similar. Online solution sites are out of bounds in this class, this is a breach of academic integrity and could result in receiving a failing grade in this class. There are plenty of resources to get help (see getting help section of this syllabus). Learning Physics is a struggle, and learning takes place in this struggle, getting solutions online does more harm than good and ultimately lowers your overall grade.

Study Groups: It is very beneficial in learning to find study partners and form study groups! Canvas will be set up with study groups under the "Peoples" in the "Groups" tab or you can use other apps to connect to your group. If you have people in the class that you want to form a study group with you can self-assign yourself to a group during the first week of class. Students can use Canvas or the app of their choice to connect with their study group. See Chapter 0 Part A lecture notes for more information on best practices for a study group.

Extra Credit: During the semester the instructor may assign problems or ask you to attend events that are not necessarily in the scope of the class, these will be counted as extra credit. They are not required and are considered optional; they will not count against your grade if you do not do them. Extra credit points are added to the total points earned on the worksheets unless the instructor specifies otherwise.

Canvas & e-mail: Get familiar with the "Canvas" web site; it is accessed through the UNF home page under "Mywings". The instructor will post course material such as lecture notes, lecture videos, practice problems, and solutions throughout the semester on this site. If you are not already using your UNF e-mail account, you should set one up, so you can access email through this account as soon as possible. This will enable you to receive important course-related information. ***Information Important to the class will be posted on Canvas so check it often!*** If you experience any problems with your UNF account, you may send an email to: helpdesk@unf.edu or call the UNF Computer Helpdesk at 904-620-4357.

Canvas is not the instructor's grade book. - Your scores on assignments will be posted on Canvas; this is only so you can view your grades in the course and an approximate grade for the class at that time but it is not the instructor grade book.

Technical Support – If you If you experience any problems with your UNF account you may send an email to: helpdesk@unf.edu or call the UNF Computer Helpdesk at 904-620-4357.

Student Learning

The student starts by learning the basic course content at home by reading the lecture notes, watching lecture videos and completing reading quizzes posted on Canvas weekly. Review the example problems in the lecture notes and work through them yourself, then review the proper problem solving techniques by watching the solving problems videos. Work through the practice problems by yourself and then meet with their study group, they get to practice using what they have learned.

For this class the instructor will be acting as your guide through Physics, the instructor will be there to help you master the concepts and clear up misconceptions through many hours of pre-recorded videos, class time, office hours, in-person and Zoom appointments.

In order to do well in this class, you must: (Learn more about succeeding in this class from the Chapter 0 lecture notes.)

1) Read the lecture notes that are posted on Canvas, they are the classes "Textbook". Read the Lecture notes to understand the material not just get through them. It is a good idea to read the notes before watching the Lecture video or have them available when watching.
2) Create a schedule for all classes and activities with time and days, block time for reading, watching videos, working problems and studying.
3) Create and attend a study group and actively participate in the group.
4) Watch the lecture videos posted on Canvas before or during the week that the material is being covered.
5) If at any point during the lecture/videos, you find difficulties. Please reach out to the instructor for help. to schedule office hours or if possible, SASS (see "Help" section of this syllabus).
6) Prepare before watching the lecture videos by reading the lecture notes, understand the basic concepts and the solved problems.
7) Come to class prepared to ask questions and actively participle in class. Understand the concepts, demonstrations and problems covered during the lecture, ask questions if you do not. Review notes and problems solved in class to reinforce what you have learned.
8) Participate in the "Class Activities" during class and small group meetings, discuss the problem with your group members, check solutions and determine the proper methods to solve the problems. Threat this time as a learning opportunity to learn what you understand and what you do not understand.
9) Review notes and problems solved in lecture notes and video solutions, to reinforce what you have learned.
10) Work through the practice problems once you have covered all the material in that chapter. Make sure you know how the problems are worked out, not that you just got the right answer. Go back over videos or notes to form a better understanding. Use the practice problems in your study groups as a tool to mastering the concepts and honing your problem-solving skills.
11) Watch the problem solving videos for proper technique and help.
12) Review graded Exams, Worksheets and In-class activities to determine what you know and areas needing work.
13) It is Important in this class to **keep up**, watch the videos, read the lecture notes, work on the practice problems, carve out time for the lab, start worksheet early and work on them over time.
14) You should set aside a minimum of **9 hours per week to just prepare for class**, this means watching the Lecture videos, reading the lecture notes, and reviewing lecture notes/problems, working practice problems and completing assignments.

Student Assessments

In order to assess how well you are learning the material and learning objectives in this class, reading quizzes, worksheets and 7 exams will be given during the semester and a cumulative final exam. Scores on theses assessments will reflect your mastery of the course material and objectives. The specific day of the week they are due is shown below, any changes to this day will be announced.

Interactive lessons (5 %) due Sunday: The Interactive Lessons will be computer simulations involving the concepts that are being covered in class. They will be in the form of a worksheet with a set of questions to be answered as you interact with the simulation. The simulations are a way to develop a deeper understanding of the Physics we are covering and to see it in a "real world" setting. The worksheet will be uploaded to the Canvas assignment.

Attendance to Lecture Class (5 %): At the end of the semester the student must have attended at least 70 % of the classes or they will receive a zero for attendance. Attendance will be taken using Canvas and a code given during the class.

Attendance > 70%	Grade for Class Attendance 100 %
Attendance < 70 %	Grade for Class Attendance 0 %

Quizzes (10 %) due Sunday: Quizzes of various types will be given during the semester to check your mastery of the material and demonstrate your understanding of the Physics. They are Canvas quizzes and are graded when taken.

Worksheets (15%) due Wednesday: The Worksheets will be "take-home" and consist of problems and questions about the reading assigned for the class modules, lecture notes and practice problems. The worksheet will be posted on Canvas and will be in the form of multiple-choice questions as one assignment and work out problems as a second Canvas assignment that the solutions are uploaded. The worksheets are in two parts;

Part A of the worksheet is short answer and multiple-choice questions that cover mostly concepts and the basic math covered in the chapters. This part of the worksheet is graded by Canvas and the student gets immediate feedback.

Part B of the worksheet are questions that require multiple steps, multiple concepts, and more extensive mathematical steps to work out the solution. The solutions to these problems are uploaded to Canvas and hand graded based on the method and process the students have developed in problem solving. These uploaded solutions will be graded by the "Problem Solving" rubric in Chapter 0 Par B – Physics Problem Solving.

All the work done on the worksheet will be scanned as one PDF document and uploaded to a Canvas Assignment, this can be a scanned paper copy of the worksheet itself if all work is done on the actual worksheet, work done on scratch paper or annotated on the worksheet using a tablet (make sure the annotations upload). All work must be shown on each of the "upload" problems, or NO credit will be given.

Exams (50%) on assigned Fridays: There will be 7 Exams given during the semester and a cumulative final during finals week. The Exams will be concept and mathematical problems. They will be similar to the Quizzes, practice problems and worksheets. The instructor reserves the right to change the format of the Exams and will give prior notice if changed.

- ONLY pencil and calculator will be allowed while taking the Exam, no other papers or devices should be present and if seen could result in a grade of zero for that Exam. Scratch paper will be provided for the Exam.
- You will have one entire 50-minute class period to complete the Exam.
- All work including equations, algebra, calculus and the final equations in symbols must be shown in your solutions to receive credit. Just having the correct numerical answer is not enough.
- If necessary, a restroom break of no more than 5 minutes can be used during the exam, any longer than this the exam will be collected and the resulting score will be zero.

Equation Sheet - The instructor will post a copy of the equation sheet on Canvas a few days before the Exam so you will be aware of what will be on the Exam. This equation sheet will be attached to the in-class Exam. Do NOT bring this equation sheet to class.

- **The lowest score of the 7 in semester exams will be dropped.**
- **The final exam is cumulative and will NOT be dropped.**

There are NO make up exams given in this class. A student who has a documented emergency/sickness or an unforeseeable absence from an exam should contact the instructor as soon as possible, normally within 24 hours after the missed exam. For an absence to be approved, documentation of the reason for absence must be provided. If the unforeseeable absence is excused by the instructor, the student will be expected to take the makeup exam. The makeup exam will cover the same subject matter as the regular exam and in a similar format, although the exams will not be identical. Students are only allowed to make up one excused exam during the semester, a score of zero will be given on any unexcused exam missed.

Final Exam (15 %) – The final exam will be given during finals week and will include all materials covered during the semester. The final exam is mandatory, must be taken, will not be dropped and will not replace any other exam!

Turning in late Assignments: To give extra incentive for students to keep up but knowing they have other responsibilities, assignments will be accepted after the due date under the following conditions.

1) Late assignments for worksheets and class activities will be accepted within two weeks from the due date, with points taken off for being late. It is the student's responsibility to notify the instructor.
2) If the solutions to the assignment have been posted the maximum points on the assignment is 50%.
3) If there is a documented emergency, and the instructor is notified before the two weeks is up the assignments will be accepted, and no late points will be deducted.

- **NO assignment (Quizzes, Labs, Class Activities or Worksheets) will be accepted two weeks after the due date and the result will be a zero for that assignment.**
- NO Assignments will be accepted after the last day of class.

Feedback from the course assessments

A grade will be assigned to each reading quizzes, worksheets. activities and Exams completed by the student, this grade will have been adjusted so that it reflects your actual grade and the weighted average (see below) of these assessments indicate your grade at the particular time in the class. There will NOT be any adjustment of your final grade at the end of the semester. All assessments will be graded, and the grade posted within two weeks of completion date or announced otherwise.

Grades: Categories and their weights which contribute to your total class score are as follows:

Average of 7 Exams (lowest dropped)	50%
Final Exam	15%
Worksheets	15%
Quizzes	10 %
Interactive Lessons	5 %
Attendance	5 %
	100%

The initial percentile ranges for determining final grades are as follows:

93.0 to 100.0	A
90.0 to 92.9	A-
87 to 89.9	B+
83.0 to 86.9	B
80.0 to 82.9	B-
75.0 to 79.9	C+
69.0 to 74.9	C
58 to 68.9	D
Below 57.9	F

Important: Your final Worksheet grade and Reading quiz average cannot exceed 105 % due to extra credit.

Your grades throughout the semester will have been adjusted and reflect your grade at that point in time, there will be NO final adjustment to your grade at the end of the semester!

The instructor reserves the right to adjust scores on assignments to achieve a particular passing grade. If made, such changes will be announced in class. A student who does not turn at least one item for grading, worksheets or quiz will be recorded at the end of the term as having never attended this class. Assigned grades, including the final grade for this course may be adjusted for academic misconduct.

Other Information for the Class

Transmission of Grades: As per the instructor's legal and ethical responsibilities to discuss a student's grades only with that student, The instructor will not discuss individual grades via e-mail or the phone. You must view your grades on Canvas or see the instructor in person or on Zoom to discuss the specifics of your grade. The student handbook outlines procedures for students wishing to appeal academic decisions.

E-mail Instructor: The preferred method of communication for the instructor is through e-mail and students can expect a response to E-mails sent to the instructor within 24 hours of being sent or two business days. The instructor will notify students when they are unavailable for an extended period of time and not able to respond in a timely manner.

Academic Integrity information: Any student found engaging in academic misconduct (as defined by the University of North Florida) will be subject to appropriate disciplinary action. The current student handbook contains a detailed discussion of the University's policy on academic misconduct. Violations of academic integrity include cheating, fabrication or falsification of information or documents, plagiarism, abuse of academic materials, and complicity in academic misconduct. Students may be subject to any and all of the following: academic counseling, reprimand, a failing grade for the assignment that involved misconduct, a failing grade for the course, and referral of the behavior to the Departmental Chair or appropriate Dean. The student handbook outlines procedures for students wishing to appeal the academic decisions of the Professor.

The books and resources we utilize in this course may contain unassigned chapters, articles, etc., that you will not be required to read. Our assigned materials are appropriate for the course and are consistent with legal requirements for General Education courses.

Misconduct: Under this heading the University of North Florida Student Handbook identifies several types of violations; these include but are not limited to: cheating; fabricating and falsifying information or citations; submitting the same work for credit in more than one course; plagiarizing; providing another student with access to one's own work to submit under this person's name or signature; destroying, stealing, or making inaccessible library or other academic resource material; and helping or attempting to help another person commit an act of academic dishonesty.

The University of North Florida authorizes any instructor who finds evidence of cheating, plagiarism, online solutions or other wrongful behavior that violates the University of North Florida Academic Integrity Code to take appropriate action. Possible action includes, but is not limited to, failing the student on the work in question, failing the student for the course, notifying the appropriate academic dean or Vice President for Student Affairs, and requesting additional action be taken. The consequences of a breach of academic integrity may result in an F, which is unforgivable, regardless of withdrawal status. The student handbook outlines procedures for students wishing to appeal decisions of the Professor. Please review the University policy on academic integrity: Academic Misconduct Policy at UNF: https://www.unf.edu/president/policies_regulations/02-AcademicAffairs/EnrollmentServices/2_0640P.aspx

Academic and Student Support Resources

Students with Disabilities: *Americans with Disabilities Act (ADA) Policy*. Students with disabilities who seek reasonable accommodation in the classroom or other aspects of performing their coursework must first register with the UNF Student Accessibility Services (SAS) center located in Building 57, Room 1500. SAS staff members work with students to obtain required documentation of disability and to identify appropriate accommodations as required by applicable disability laws including the Americans with Disabilities Act (ADA). After receiving all necessary documentation, the SAS staff determines whether a student qualifies for services with the SAS and, if so, the accommodation will be provided to the student. SAS staff then prepares a letter for the student to provide faculty advising them of approved accommodation. For further information, contact the SAS by phone at 904-620-2769, by email at sascenter@unf.edu, or visit SAS website at https://www.unf.edu/sas/

Military and veteran students who return from combat exposure may be utilizing the post 9/11 GI bill to continue postsecondary education goals and may need both physical and academic accommodations. Contact Ray Wikstrom, Director of Military and Veterans' Resource Center by phone (904) 620-2655, email (ray.wikstrom@unf.edu). Contact the Military and Veterans' Resource Center by phone (904) 620-5131 or email mvrc@unf.edu

Thomas G. Carpenter Library: Students are encouraged to utilize the many resources available to them in the University's library. The librarians at Carpenter Library offer one-on-one consultations with students to assist them in research projects, consult with you on finding/accessing materials, citing sources appropriately, and other scholarly activities. The library also has a Virtual Learning Center that offers students the opportunity to explore virtual reality technology.

Student Health and Wellbeing: The UNF counseling center can help students who are having difficulties managing stress, adjusting to college, or who are feeling sad and hopeless. You can reach the UNF counseling center at http://www.unf.edu/counseling-center/ or by calling (904)-620-2602 during and after business hours for routine appointments or if you or someone you know is in crisis. Walk-in hours are 10-2 PM Monday-Friday. Crisis appointments available.

The Victim Advocacy Program provides confidential crisis intervention services to anyone in the UNF community impacted by crime or victimization. Services offered range from emotional support to assistance navigating the criminal justice system. The Victim Advocacy Program is located in Founders Hall, Building 2, Suite 2100. Students may contact the 24-hour Crisis Helpline at (904) 620-1010.

Non-Discrimination Policy: The University of North Florida (UNF) is committed to providing an inclusive and welcoming environment for all who interact in our community. To accomplish this intent, UNF shall not commit or permit discrimination or harassment on the basis of genetic information, race, color, religion, age, sex, disability, gender identity/expression, sexual orientation, marital status, national origin or veteran status in any educational, employment, social or recreational program or activity it offers. Similarly, UNF will not commit or permit retaliation against an individual who complains of discrimination or harassment or an individual who cooperates in an investigation of an alleged violation of University Regulation. In exercising these standards, the University will not abridge either free speech or academic freedom based on its context. Accordingly, any member of the UNF

community who believes that they have been subjected to discrimination, discriminatory harassment, retaliation, or sexual misconduct may seek guidance, counseling and/or file a complaint by contacting: Cheryl Gonzalez, Director, E.O.D. and Title IX Administrator, located at Building One, J.J. Daniel Hall, Suite 1201, 1 UNF Drive, Jacksonville, Florida 32224-7699, or call (904) 620-2507 or via 711 Florida Relay for persons who are deaf or hard of hearing or those with speech impairments and/or limitations.

Technical Support: If you experience any problems with your UNF account you may send an email to: helpdesk@unf.edu or call the UNF Computer Helpdesk at 904-620-4357.

First Assignment: Completion **of the first quiz by the due date is critical! Failure to meet the deadline could impact your ability to receive institutional or federal aid.**

Change or disruption of class schedule.

Any changes to the above class description, assignments or the schedule will be announced in class. It is the responsibility of the student to keep abreast of any announcements made during class. In the event of disruption of normal classroom activities due to an emergency such as hurricane, pandemic or other unforeseen event or combination of events, the format of this course may be modified in order to enable completion of the course requirements. This course may be taught remotely and require online access off campus. In that event, you will be provided with an addendum to this syllabus that will supersede this version. It is your responsibility as a student participant to be proactive during any emergency to find instructions that will be posted on Canvas which you should check often. It may be necessary to switch to remote learning through Zoom online conferencing, if this should happen the instructor will notify you of the changes to the class format. This change may require adjustments to the assessment methods, assignments and the due dates for worksheets and exams. The instructor will keep you up to date with all changes to the class, check Canvas and e-mails often!

Table: Tentative Class Schedule Fall 2026

Calculus-based Physics 2 CCEC PHY 2049C CRN # 84245 Tentative Schedule for Fall 2026				
Week	**Dates of Classtime**	**Topics/Chapters Covered Each Week**	**Worksheets Due Wednesday**	**Exams**
1	17-Aug 19-Aug 21-Aug	Intorduction to class and Chapter 0 Part A and Part B Chapter V - Vector Review Chapter 34 The Electric Charge Chapter 35 The Electric Force		
2	24-Aug 26-Aug 28-Aug	Chapter 35 The Electric Force Exam Review - Class Activity	**WS 1**	**Exam 1 - Friday 8/28**
3	31-Aug 2-Sep 4-Sep	Chapter 36 The Electric Field Part A Chapte 36 The Electric Field Part B Chapter 36 The Electric Field Part C Chapter 37 Conductors		
4	7-Sep 9-Sep **11-Sep**	**No Class Labor Day** Exam Review - Class Activity	**WS2**	**Exam 2 - Friday 9/11**
5	14-Sep 16-Sep 18-Sep	Chapter 38 Electric Potential Chapter 39 More on the Electric Potential Chapter 41 Current, Resistance and Ohm's Law Chapter 42 Electrical Power		
6	21-Sep 23-Sep **25-Sep**	Chapter 42 Electrical Power Exam Review - Class Activity	**WS 3**	**Exam 3 - Friday 9/25**
7	28-Sep 30-Sep 2-Oct	Chapter 43 Electric Circuits Chapter 44 More on Electric Circuits Chapter 45 Magnetism Chapter 46 Magnetic Forces Chapter 47 Current and the Magnetic Field		
8	5-Oct 7-Oct **9-Oct**	Chapter 47 Current and the Magnetic Field Exam Review - Class Activity	**WS 4**	**Exam 4 - Friday 10/9**
9	12-Oct 14-Oct 16-Oct	Chapter 48 Electromagnetic Induction Chapter 49 Induction in Circuits Chapter 50 Electric Motors and Power Transmission Chapter 51 Applications of EMI Chapter 52 ac Circuits		
10	19-Oct 21-Oct **23-Oct**	Chapter 52 ac Circuits Exam Review - Class Activity	**WS 5**	**Exam 5 - Friday 10/23**
11	26-Oct 28-Oct 30-Oct	Chapter 53 Electro-magnetic Waves Chapter 54 Electro-magnetic Spectrum Chapter 55 EM Wave Intensity Chapter 56 EM Wave Polarization Chapter 57 Introduction to Optics		
12	2-Nov 4-Nov **6-Nov**	Chapter 57 Introduction to Optics Exam Review - Class Activity	**WS 6**	**Exam 6 - Friday 11/6**
13	9-Nov 11-Nov 13-Nov	Chapter 58 Refraction of Light **No Class Veterians day** Chapter 59 Imaging Mirrors Chapter 60 Imaging Lenses Chapter 62 Optical Instruments **Last day to withdraw from class**		
14	16-Nov 18-Nov **20-Nov**	Chapter 62 Optical Instruments Exam Review - Class Activity	**WS 7**	**Exam 7 - Friday 11/20**
15	23-Nov 25-Nov 27-Nov	**No Class - Thanksgiving Holiday (11/23 - 11/27)**		
16	30-Nov 2-Dec 4-Dec	Chapter 63 Wave Optics Chapter 64 Diffraction of Light Chapter 65 Thin film optics	**WS 8**	
17		**Final Exam Week**	**Final Exam Wednesday 12/9 at 7:00 am**	**Chapter 34 - 65**

Any changes to the above schedule will be announced in class prior to the change.
Exam dates rarely change but the chapters they cover may be revised by the instructor depending on what is covered in lecture. Notice of any change will be given before the exam.